HMDS FOR THE METAVERSE

Are we really there yet?

Mithun Mohandas | mithun@digit.in

While the concept of the Metaverse has been around in science fiction for decades, it's only in recent times that we've seen the biggest tech giants in the world take concrete steps towards realising a massive interconnected digital world with strong ties to the physical world. We've seen different ways of how the Metaverse could finally pan out in movies and other media - Ready Player One, Surrogates and The Matrix are a few examples that come to mind. However, these dystopian depictions feel like the concept of the Metaverse has been stretched to absurd levels. Perhaps, we might end up like little batteries or be cooped up in our homes for months on end, or perhaps life might turn out completely different. It certainly seems like the concept of Metaverse will end up being a reality one way or the other. So how are we going to plug into the Metaverse?

SMARTPHONES

The answer that everyone jumps to is not quite the easiest one. The first commercially available VR HMD (Head-Mounted Display) was a bulky monstrosity that needed to be connected to a powerful computer. The HMD would render the visuals while the PC would churn out what needed to be rendered. Both the devices required to connect to the Metaverse are significant investments. This was

back in 2014. Over time, VR HMDs have become leaner and the dependence on a powerful PC has gone away. Current HMDs even sport powerful mobile SoCs which are complete solutions that feature significant compute power and even have the network bandwidth to ensure a seamless experience. If not rely on the HMDs themselves for churning out the visuals, you could simply slide your smartphone into a frame and you'd have an HMD in no time. After all, everyone has a smartphone, right? Even if you had doubts about whether a smartphone SoC can handle the compute demands of rendering a virtual world, there are even remote compute services that will render life-like visuals on a server far away and beam the final visual straight to your HMDs. So it would seem that the bulkiness and the compute power constraints have been taken care of. And yet, we have giants like Google who wish to differ.

"We saw a lot of potential in smartphone VR—being able to use the smartphone you carry with you everywhere to power an immersive on-the-go experience," Google's spokesperson explained to Variety in 2019. "But over time we noticed some clear limitations constraining smartphone VR from being a viable long-term solution. Most notably, asking people to put their phone in a headset and lose access to the apps they use throughout the day causes immense friction."

Yes, the smartphone is indeed very powerful but it's also a device that already serves a different purpose. Unless the user experience is crafted in such a manner that the smartphone's functionalities are still accessible even when placed into an HMD, there's little chance that the smartphone could end up being the enabler of the Metaverse. Moreover, smartphones by virtue of their form factor cannot extend themselves to provide a holistic VR experience. You can't expect the smartphone to have the right screen resolution and refresh rate for ensuring a good experience. At least, not in the near future. And you certainly cannot expect a smart-